

Core Module Testing Tools

Core Module Testing Tools

1. Course Content
2. Starting the Dify Service
3. Enter the test tool path
4. Test the local speech synthesis function
5. Test Local Speech Recognition Functionality
6. Testing Robot Access to Dify
7. Online Voice Service
 - 7.1 iFlytek Speech Recognition Service
 - 7.2 iFlytek Speech Synthesis Service

1. Course Content

- multi_brains provides minimal test programs for testing the core modules of a large model, used to quickly locate problems in abnormal situations.

2. Starting the Dify Service

Note: If using a 2025 image, you need to copy the test folder from the "Source Code" folder to your current board for testing. For Jetson Nano and Raspberry Pi 5 boards, you need to operate within Docker. The path is

```
cd ~/yahboom_ros2_ws/yahboomcar_ws/src/multi_brains/test
```

Replace the text folder in the path with the text folder from the "Source Code" folder.

- Connect to the car's infotainment system via VNC or SSH. Enter the following command in the terminal:

```
sh ~/bringup_dify.sh
```

```
jetson@yahboom: ~  
ROS_DOMAIN ID: 62 | ROS: humble  
my_robot_type: M1 | my_lidar: c1 | my_camera: usb  
-----  
jetson@yahboom:~$ sh ~/bringup_dify.sh  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "CERTBOT_EMAIL" variable is not set. Defaulting to a blank string.  
.  
WARN[0000] The "CERTBOT_DOMAIN" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
[+] Running 10/10  
✓ Container docker-sandbox-1      Runni...      0.0s  
✓ Container docker-web-1          Running       0.0s  
✓ Container docker-ssrf_proxy-1   Ru...        0.0s  
✓ Container docker-db-1           Healthy      0.5s  
✓ Container docker-redis-1        Running       0.0s  
✓ Container docker-worker_beat-1   R...         0.0s  
✓ Container docker-plugin_daemon-1 Running       0.0s  
✓ Container docker-api-1          Running       0.0s  
✓ Container docker-worker-1        Runnin...    0.0s  
✓ Container docker-nginx-1        Running       0.0s  
jetson@yahboom:~$
```

3. Enter the test tool path

```
cd yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/test/  
ls
```

- The test tool is as follows:

```
jetson@yahboom:~$ cd yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/test/  
jetson@yahboom:~/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/test$ ls  
test_baiduTTS.py      test_pep257.py        test_TongyiTTS.py  
test_copyright.py     test_PiperTTS.py      test_xunfeiASR.py  
test_dify_connection.py test_SenseVoiceSmall.py test_xunfeiTTS.py  
test_flake8.py         test_TongyiASR.py  
jetson@yahboom:~/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/test$
```

4. Test the local speech synthesis function

- Test the local speech synthesis function for Chinese and English in sequence.

```
python3 test_PiperTTS.py
```

After running, it will play the Chinese and English speech synthesis results sequentially.

```
jetson@yahboom:~/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/test$ python3
test_PiperTTS.py
pygame 2.6.1 (SDL 2.28.4, Python 3.10.12)
Hello from the pygame community. https://www.pygame.org/contribute.html
测试中文语音合成
Test English speech synthesis
```

5. Test Local Speech Recognition Functionality

- After running, it will test the preset Chinese and English audio recordings sequentially.

```
python3 test_SenseVoiceSmall.py
```

After running, it will print the speech recognition results of the test audio.

```
jetson@yahboom:~/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/test$ python3
test_SenseVoiceSmall.py
=====测试本地中文语音识别=====
=====
funasr version: 1.2.6.
SenseVoiceSmall ASR_Engine initialized successfully
rtf avg: 0.425: 100%|████████████████████| 1/1 [00:00<00:00, 1.95it/s]
听见啦
=====Test local English speech recognition=====
=====
funasr version: 1.2.6.
SenseVoiceSmall ASR_Engine initialized successfully
rtf avg: 0.076: 100%|████████████████████| 1/1 [00:00<00:00, 5.79it/s]
okay i'm here
jetson@yahboom:~/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/test$
```

6. Testing Robot Access to Dify

- When the program displays "Model service unavailable," and it's unclear whether the robot's inability to access Dify is due to a network address error, you can test:

```
python3 test_dify_connection.py
```

If the robot displays the following information normally, it proves that the robot's connection to Dify is normal. Otherwise, it may be that Dify is not started or the Dify program has crashed and needs to be restarted.

```
jetson@yahboom:~/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/test$ python3
test_dify_connection.py
/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/config/multi_brain
ns_setting.yaml
Connection dify successful
jetson@yahboom:~/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/test$
```

7. Online Voice Service

7.1 iFlytek Speech Recognition Service

- If online speech recognition encounters errors, you can test iFlytek's speech recognition service separately.

```
python3 test_xunfeiASR.py
```

- After running, it will perform speech recognition tests on the preset test audio.

```
jetson@yahboom:~/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/test$ python3 test_xunfeiASR.py
=====test Xunfei XUNFEI_ASR =====
Xunfei ASR_Engine initialized successfully
Here we go.
jetson@yahboom:~/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/test$
```

7.2 iFlytek Speech Synthesis Service

- If online speech synthesis encounters errors, you can test iFlytek's speech recognition service separately.

```
python3 test_xunfeiTTS.py
```

After running, it will first perform speech synthesis on the default text, and then play the corresponding audio.

```
jetson@yahboom:~/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/test$ python3
test_xunfeiTTS.py
pygame 2.6.1 (SDL 2.28.4, Python 3.10.12)
Hello from the pygame community. https://www.pygame.org/contribute.html
test Xunfei TTS_Engine
Xunfei TTS_Engine initialized successfully
jetson@yahboom:~/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/test$
```